

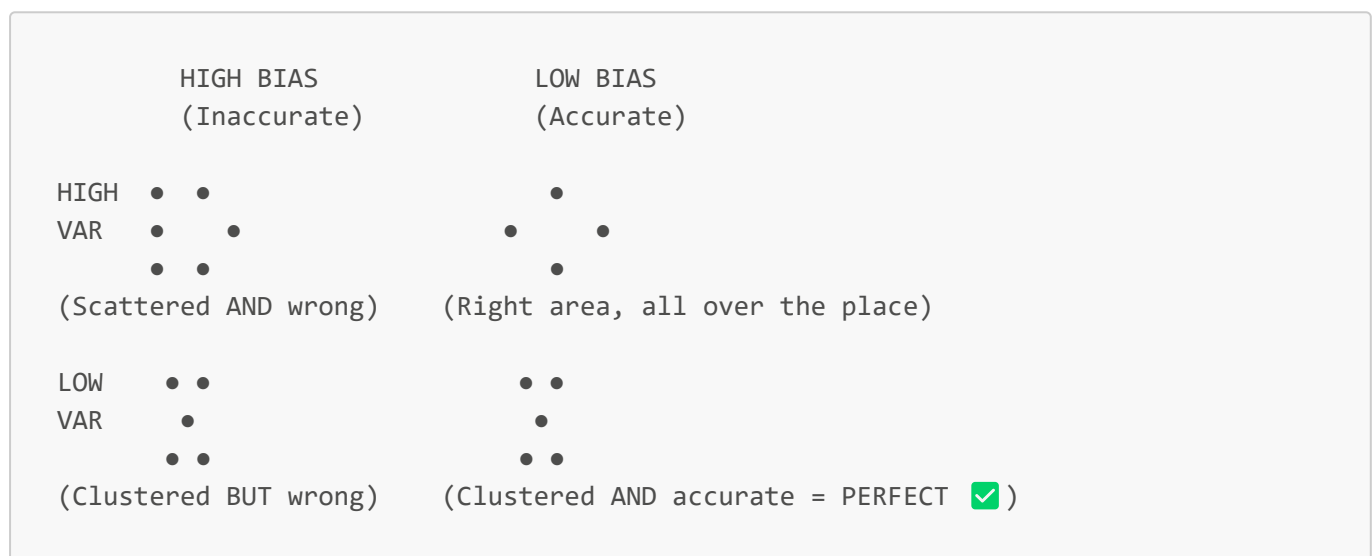
# Bias, Variance, Underfitting & Overfitting — Q&A Notes

A permanent mental model for the most fundamental tradeoff in all of machine learning

## The Archery Analogy

### Q: What is the most intuitive way to understand bias and variance together?

Imagine training an archer (your model) to hit the bullseye (the true real-world pattern). Each arrow is a prediction on a new data point:



| Combination               | Meaning                           | Real Name         |
|---------------------------|-----------------------------------|-------------------|
| High Bias + High Variance | Scattered everywhere AND wrong    | Worst case        |
| High Bias + Low Variance  | Consistently wrong, same spot     | Underfitting      |
| Low Bias + High Variance  | Right area but all over the place | Overfitting       |
| Low Bias + Low Variance   | Bullseye every time               | <b>The Goal</b> ✓ |

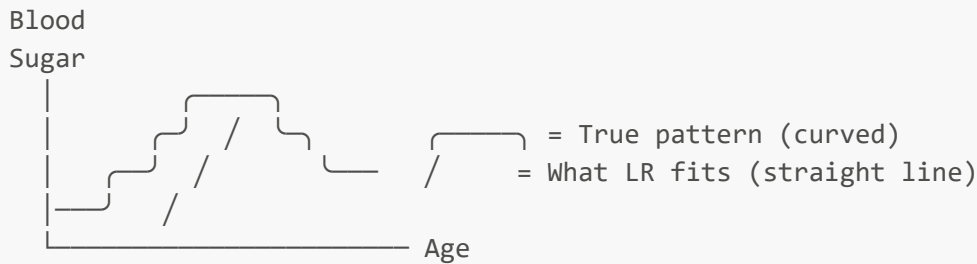
## What Is Bias?

### Q: What exactly is bias, and where does it come from?

Bias is the error that comes from **wrong assumptions in your model**.

It answers: *"Is your model even capable of learning the true pattern?"*

Suppose the true relationship between Age and Blood Sugar is a curve, but you force Linear Regression (a straight line) to learn it:



The straight line can **never** capture the curve no matter how much data you give it. This gap between what the model can learn and the true pattern is Bias — a fundamental limitation of the model's assumptions.

**High Bias = Model is too simple = Underfitting**

## What Is Variance?

**Q: What exactly is variance, and where does it come from?**

Variance is how sensitive your model is to the **specific training data it saw**.

It answers: "If I trained on a slightly different set of patients, would I get a wildly different model?"

Imagine training KNN with K=1 on two slightly different samples of 800 patients:

Training Set A → Model A draws:



Training Set B (just 5 different patients) → Model B draws:



COMPLETELY different boundaries from nearly the same data!

This wild swinging from one training set to another is Variance. The model is fitting **noise**, not the true pattern.

**High Variance = Model is too complex = Overfitting**

## Underfitting vs. Overfitting

**Q: What does underfitting look like in practice on a real dataset?**

Using BMI → Diabetic? (D = Diabetic, N = Not Diabetic):

|       |    |    |    |    |    |    |    |    |    |
|-------|----|----|----|----|----|----|----|----|----|
| BMI:  | 20 | 22 | 25 | 28 | 30 | 33 | 35 | 38 | 40 |
| True: | N  | N  | N  | N  | D  | D  | D  | D  | D  |

```
Model: N   N   N   N   N   N   N   N   N   N ← predicts everyone Not Diabetic
```

The model is so simple it learned nothing useful — just predicts the majority class.

```
Training Accuracy: 55% ← bad
Test Accuracy:     54% ← also bad, consistently bad
```

Both training and test accuracy are **low and close together**. The model went to the exam without studying.

### Q: What does overfitting look like in practice on a real dataset?

The training data had noise — a BMI=25 patient happened to be diabetic due to genetics; a BMI=33 patient happened to be non-diabetic due to medication. The model memorized these noise points:

|        |    |    |               |    |    |               |    |    |    |
|--------|----|----|---------------|----|----|---------------|----|----|----|
| BMI:   | 20 | 22 | 25            | 28 | 30 | 33            | 35 | 38 | 40 |
| True:  | N  | N  | N             | N  | D  | D             | D  | D  | D  |
| Model: | N  | N  | D             | N  | D  | N             | D  | D  | D  |
|        |    |    | ↑             |    |    | ↑             |    |    |    |
|        |    |    | Noise point!  |    |    | Noise point!  |    |    |    |
|        |    |    | Memorized it. |    |    | Memorized it. |    |    |    |

```
Training Accuracy: 99% ← looks amazing!
Test Accuracy:     61% ← falls apart on new patients
```

Training accuracy is **very high**, test accuracy is **much lower**. The model crammed everything including the wrong stuff — it aced the practice paper but failed the real exam.

### Q: What does the sweet spot between underfitting and overfitting look like?

```
Training Accuracy: 87% ← not perfect, but real
Test Accuracy:     85% ← close to training = generalizes!
Gap = 2%           ← small gap is the sign of a good model
```

The model learned the true pattern without memorizing the noise. Both accuracy values are high **and close to each other**.

## The Bias-Variance Tradeoff

### Q: What is the mathematical decomposition of total model error?

$$\text{Total Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Noise}$$

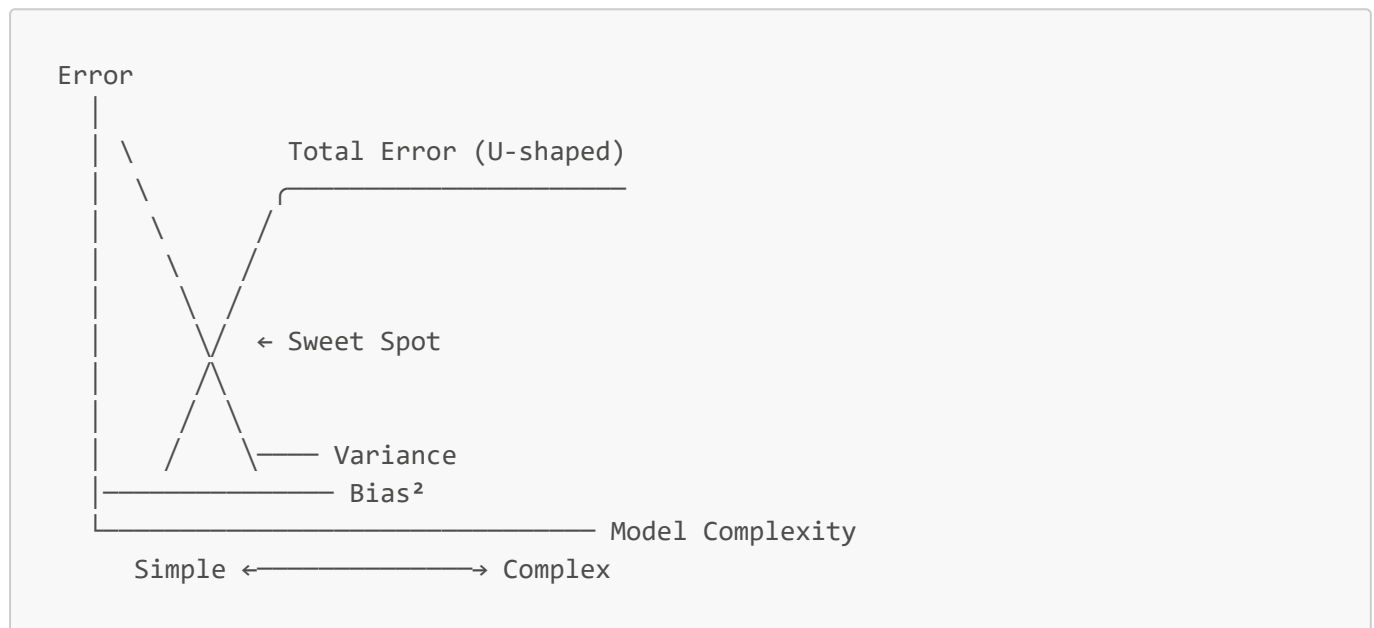
**Irreducible noise** is the randomness in the real world that no model can ever eliminate — e.g., a perfectly healthy person randomly develops diabetes with no predictive features. This floor cannot be removed no matter how good your model is.

### Q: Why can't you reduce both bias and variance at the same time?

As you increase model complexity:

- **Bias decreases** — the model becomes capable of learning more complex patterns
- **Variance increases** — the model becomes more sensitive to the specific training data

They move in **opposite directions**. Every step to fix one automatically worsens the other:



Tuning is the act of **finding the lowest point of the total error curve**.

## How Each Model Lives on This Spectrum

### Q: Where do Linear Regression, Logistic Regression, SVM, and KNN sit on the bias-variance spectrum?

| ← HIGH BIAS<br>(Underfitting)              |                                              |                        | HIGH VARIANCE →<br>(Overfitting) |              |
|--------------------------------------------|----------------------------------------------|------------------------|----------------------------------|--------------|
| Linear<br>Regression<br>(no<br>regulariz.) | Logistic<br>Regression<br>(no<br>regulariz.) | SVM<br>(RBF<br>kernel) | KNN<br>(K=7)                     | KNN<br>(K=1) |

And hyperparameters **slide each model along this spectrum**:

|     |           |            |               |
|-----|-----------|------------|---------------|
| LR: | C=0.001   | C=1        | C=1000        |
|     | High Bias | Sweet Spot | High Variance |

KNN: K=500 ————— K=11 ————— K=1  
 High Bias                      Sweet Spot                      High Variance

SVM: Low C, Low  $\gamma$  ————— Right C,  $\gamma$  ————— High C, High  $\gamma$   
 High Bias                      Sweet Spot                      High Variance

## Diagnosing the Problem

**Q: How do you diagnose whether your model is underfitting or overfitting in practice?**

Just look at two numbers — Training Accuracy vs. Test Accuracy:

```

Training Accuracy    vs    Test Accuracy

Both LOW?
→ HIGH BIAS → Underfitting
→ Fix: Use a more complex model or add more features

Training HIGH, Test LOW?
→ HIGH VARIANCE → Overfitting
→ Fix: Regularize, get more data, reduce complexity

Both HIGH and CLOSE to each other?
→ LOW BIAS + LOW VARIANCE → Sweet spot ✅

Training LOW, Test HIGH?
→ Impossible in practice. Check your data split.
```

| Pattern              | Diagnosis                   | Fix                                      |
|----------------------|-----------------------------|------------------------------------------|
| Both low             | High Bias — Underfitting    | More complex model, add features         |
| Train high, Test low | High Variance — Overfitting | Regularize, more data, reduce complexity |
| Both high and close  | Sweet spot ✅                | Nothing — ship it                        |
| Train low, Test high | Data split error            | Re-examine your pipeline                 |

## The One-Paragraph Mental Lock

**Q: What is the single paragraph that permanently locks in the entire concept?**

Bias is how wrong your model's core assumptions are — a straight line trying to fit a curve will always be wrong, no matter how much data you give it. Variance is how nervous your model is — a K=1 KNN changes its entire mind if you swap just 5 training patients. Underfitting is high bias: the model went to the exam without studying. Overfitting is high variance: the model memorized the textbook word-for-

word including the typos, so it fails on the real exam. The tradeoff exists because every step you take to fix bias (make the model more complex) automatically increases variance, and vice versa. Cross-validation is your instrument to measure where you are on this curve. Hyperparameter tuning is your steering wheel to move toward the sweet spot. The goal is always the same: **the smallest possible gap between training and test accuracy, with both being high.**

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